Vincent Chen

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EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge — *Bachelors of Science*

- **S.B.** degree in Course 18 (Mathematics) + Course 6-3 (Computer Science and Engineering). **GPA:** 5.0/5.0.
- **Relevant Coursework:** Fundamentals of Programming (6.1010), C/Assembly (6.1903), Algebra I (18.701), Real Analysis (18.100B).

Stuyvesant High School, New York — *High School Diploma*

- **GPA:** 98/100. **SAT:** 1580. 5s on **15 APs** taken. Awarded New York State Scholarship for Academic Excellence **(top 1% GPA)**.
- **Relevant Coursework:** Multivariable Calculus, Cybersecurity, Systems-Level Programming, AP Physics C, Organic Chemistry I/II, Advanced Chemistry Lab, Math Team. **Self-Studied:** Abstract Algebra, Linear Algebra, Inorganic + Physical Chemistry.
    - High Honors in Computer Science, Organic Chemistry; Terra NYC STEM Finalist; Computer Science Service Award.
- **Extracurriculars:** President of: Computer Science Dojo, Math Survey, Stuyvesant Alumni Mentoring Program, ACS Chemistry.

WORK HISTORY

NYC Dept. of Parks & Recreation — *Management & Operations Intern* (Jun 2024 – Aug 2026)

- Led projects worth **$70,000+** and managed **700+ volunteers** over **1,200+ man hours** in Lower Manhattan parks.
- Spearheaded the IPaRC app for widespread use in **300+ projects** (see below) by Parks staff and volunteer organizations.
- Honed skills in leadership, GIS, public speaking, and emergency management with **20+ FEMA Certifications** in Management/Project Planning. Awarded **Gold Presidential Volunteer Service Award** for my management and oversight.

American Chemical Society @ Rutgers University Newark — *Research Intern* (Apr 2025 – Aug 2026)

- Synthesized HfO_2-doped carbon nanotube films to investigate ferroelectric properties, with **33-fold** increase in H-diffusion.
- Designed/executed experiments involving scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDXS), and data analysis via **Excel/Python** and chronopotentiometry to evaluate catalytic morphology and activity.
- Collaborated with an interdisciplinary team to investigate multifunctional oxide materials with energy-related applications, presenting findings at **ACS Fall 2025 National Conference.**

PROJECTS

Arbiter Chess Engine (Jul 2026 – Aug 2026)

- Devised a Negamax algorithm with alpha-beta pruning to evaluate legal chess positions, utilizing **C++** for move generation.
- Built a UCI protocol handler to parse position/search commands from chess GUIs, with live terminal play against the engine.
- Link: https://github.com/cv2630/arbiter

Personal Website (Nov 2025 – Jul 2026)

- Designed a personal website for personal use, project tracking, and blogging, using **HTML, CSS**, and **JavaScript** elements.
- Link: http://cv2630.github.io

Improvement in Parks and Repair Considerations (IPaRC) (Jun 2024 – Aug 2024)

- Developed a horticulture webapp (IParC) using **ArcGIS Pro & Python**, mapping **300+** work sites in Manhattan's **200+ parks**.
- Streamlined project tracking and aided data-driven planning for NYC Parks staff, volunteers, and corporate sponsors.
- Link: https://nycp-personal.maps.arcgis.com/apps/webappviewer/index.html?id=8c23b9f4ee4b40a7bae728e9b52aec9a

AWARDS

- **Regeneron STS Scholar**: Ranked top 300/2,612 applications; awarded $2,000 for research on HfO_2@CNT membranes.
- **4x AIME Qualifier**: Ranked top 5% in olympiad math, qualified 4x for American Invitational Mathematics Examination (AIME).
- **ACS Catalyst Scholar**: Received a $40,000 scholarship toward undergraduate studies in science/engineering.
- **2x USAMTS Winner**: Awarded twice by the USA Mathematical Talent Search, a proof-based contest sponsored by the NSA.
- **USACO Silver**: Advanced to Silver (15%) in USA Computing Olympiad.

SKILLS

Languages: Python, C, C++, HTML/CSS, Java, JavaScript, Rust, SQL. **Tools/Libraries:** NumPy, Scikit-learn, Pandas, Git.

Concepts: Algorithms, Data Structures, OOP, Cryptography, Memory Management, Networking, Generative AI, Linux, MacOS.